



LUNARMARK

Lunar Surface Prospecting & Data Services

Preliminary Business Plan · March 2026

Vision

Build the authoritative ground-truth dataset of the lunar south pole — the foundational data infrastructure every mining, science, and human mission will depend on.

Why Now

SpaceX Starship heavy cargo lunar delivery is on track for 2027-2028. The window between launch capability existing and operational infrastructure being built is where foundational companies are founded.

1. Executive Summary

LunarMark is a lunar surface prospecting and data services company. The company deploys robotic assets to permanently shadowed regions (PSRs) of the lunar south pole to collect high-resolution ground-truth data on water ice concentration, regolith composition, thermal gradients, and topography — then delivers that data as a service to NASA, national space agencies, and the emerging commercial lunar sector.

THE CORE INSIGHT

Every entity planning lunar resource extraction, human habitation, or south pole operations needs ground-truth data that does not yet exist. Orbital instruments have fundamental resolution limits. No mission has put sensors inside the cold traps at scale. LunarMark does that first — and owns the dataset. This is the Planet Labs model applied to the moon.

Metric	Detail
Company Name	LunarMark Inc. (working name)

Business Model	Robotic surface prospecting + data-as-a-service
Primary Customers	NASA, ESA, JAXA, commercial lunar operators, He-3 extractors
Near-Term Revenue	NASA SBIR grants, CLPS payload contracts, data licensing
Market Size	\$11.4B lunar exploration tech (2025) growing to \$21.4B by 2030 at 13.3% CAGR
Funding Required	Seed: \$2-4M; Series A: \$15-25M post-first-flight
Key Milestone	First payload on lunar surface by 2029; first PSR interior data by 2030

2. Problem & Opportunity

The Data Gap

Water ice has been confirmed at the lunar south pole by multiple missions. LCROSS impact analysis showed approximately 6% water concentration in some areas. LRO data published in October 2024 showed ice deposits more widespread than previously thought, extending to at least 77 degrees south latitude. However, what is confirmed from orbit is insufficient for what operational and commercial missions require next.

Orbital datasets have fundamental limitations that no ground mission has addressed at scale:

- Resolution ceiling: orbital instruments resolve at 10-100m scale. Mining operations require centimeter-to-meter resolution.
- Depth uncertainty: neutron spectrometers detect hydrogen to approximately 0.5m depth. Concentration profiles at 1-3m — the practical extraction zone — remain unknown.
- Form factor unknown: whether ice exists as discrete chunks, thin films, or chemically bound material determines the entire extraction technology stack.
- Thermal mapping gaps: temperature gradients inside PSRs are modeled, not measured. Equipment survival in PSR environments depends on accurate surface data.
- NASA's VIPER rover was cancelled in 2024 after \$450M in development, leaving a validated scientific need with no mission to answer it.

THE OPPORTUNITY

Whoever builds the authoritative, high-resolution ground-truth dataset of the lunar south pole becomes structurally indispensable to every subsequent mission. The data compounds in value as the lunar economy grows and cannot be replicated without a new mission costing tens of millions of dollars.

Market Drivers

- NASA Artemis program targets human south pole landing 2027-2028, requiring detailed site characterization in advance.
- CLPS program extended to large payloads post-2025; most landing sites target the south pole specifically.
- SpaceX Starship cargo variant capable of delivering up to 100 metric tons to the lunar surface — unlocking large-scale operations requiring detailed advance planning.
- China's lunar program targets a crewed south pole mission before 2030, creating geopolitical urgency for US and allied data advantages.
- At least five commercial entities are actively planning lunar resource extraction — all require prospecting data before committing capital.
- Interlune announced a 2025 DOE agreement to deliver helium-3 harvested from lunar soil to Earth by 2029, creating an additional high-value customer segment.

3. Solution

What LunarMark Builds

LunarMark develops and deploys a two-platform robotic system optimized for the lunar south pole environment. The platforms serve complementary roles: broad-area surface survey at speed, and precision instrument access to PSR interiors. The core product is not the hardware — it is the data collected and the analytics platform through which customers access it.

System	Description	Customer Value
IMPASS-Derived Survey Platform	Active spoke wheel robot based on RoMeLa/UCLA research lineage; individually actuated spokes for terrain adaptation; distributed spoke-tip sensors for continuous transect data collection across wide survey areas	Fast broad-area mapping of surface temperature, regolith hardness, and spectral signatures — dense continuous data rather than point measurements
Polar Hopper	Cryogenic-rated semi-autonomous robot; tether-powered from rim station; drill, mass spectrometer, ground-penetrating radar, thermal sensors; deployed for precision PSR interior access	Direct physical measurement inside cold traps — not achievable from orbit or non-PSR surface positions
Rim Station	Solar-powered hub on crater rim; power relay, communications	Enables continuous operations without nuclear power licensing;

	uplink, data storage; Shackleton Crater rim receives sunlight approximately 89% of the time	powers both surface platforms
Data Platform	Cloud API and visualization tool; processed data products including ice concentration maps, depth profiles, thermal models, and topographic surveys	Actionable intelligence in standard formats immediately usable by mission planners and operations teams
Custom Surveys	Commissioned prospecting of specific target zones for customers planning landing sites or resource claims	Directly reduces mission risk; justified at significant premium over standard data licensing

Proposed Locomotion: Active Spoke Wheel Platform

The broad-area survey platform draws on two decades of academic research into rimless wheel locomotion, specifically the IMPASS (Intelligent Mobility Platform with Active Spoke System) architecture developed by Dr. Dennis Hong's Robotics and Mechanisms Laboratory (RoMeLa) at UCLA, with published kinematic analysis extending back to 2005 ASME Mechanisms and Robotics conferences.

Design Principle

IMPASS uses rimless wheels with individually actuated spokes rather than a continuous rim. Each spoke acts as both a leg and a contact sensor — the robot rolls across terrain by vaulting sequentially from spoke tip to spoke tip, with the hub tracing a cycloidal path above the surface. The active element is critical: each of the nine degrees of freedom allows individual spoke extension and retraction in real time, enabling the platform to adapt its effective geometry to terrain features as it moves.

The result is a locomotion system that combines the mechanical simplicity of a wheel with the terrain adaptability of a legged robot — stepping over obstacles larger than its spoke length, following surface contours, and maintaining forward momentum across highly irregular ground without the actuator count or control complexity of a full bipedal or quadrupedal system.

Why This Architecture Fits the Lunar Environment

Property	Lunar Relevance
Minimal rim contact	No continuous grinding surface exposed to abrasive regolith; spoke tips make brief, discrete contacts reducing dust infiltration opportunity
Individual spoke actuation	Terrain adaptation without complex gait computation; each spoke responds to its own contact sensor independently
Low parts count per	Fewer failure modes than legged alternatives in a no-repair, high-

degree of terrain adaptation	radiation, cryogenic environment
Natural low-gravity gait	In 0.16g, the vaulting motion between spoke contacts produces extended ballistic phases — the platform effectively skims the surface, covering ground quickly at low energy cost
Distributed tip contact sensors	12 foot touch sensors in the original design are directly repurposable as science instruments: thermal probes, resistivity sensors, or regolith hardness measurements collected passively during transit
Stable rest configuration	Can lock into a tripod stance on three spoke tips — provides a stable sensor platform for precision measurements without requiring a separate deployment mechanism

Operational Role: Survey Before Precision

The IMPASS-derived platform serves as the first-pass survey instrument. It covers wide areas quickly — generating continuous transect data on surface temperature, albedo, and regolith properties — and identifies high-priority targets for the precision Polar Hopper. This two-stage architecture means the expensive, slow precision measurements are directed by fast broad survey data rather than deployed blindly, dramatically increasing the science return per mission hour.

A single spoke-wheel platform traversing a 500m transect across the crater rim approach zone at 0.5 m/s collects roughly 1,000 discrete surface contact measurements per hour. At three contact points per spoke cycle across a 3-spoke hub configuration, that represents continuous characterization of the surface at sub-meter resolution — a data density unachievable by any current lunar rover design.

Development Path

LunarMark does not need to develop this locomotion system from scratch. The kinematic analysis, gait planning mathematics, and control architecture are extensively published in peer-reviewed literature. The development task is adaptation: reparametrizing gait models for 0.16g, qualifying actuator and sensor systems for cryogenic operation, dust-proofing the joint interfaces, and integrating science payload into the spoke tips. A research collaboration with Dr. Hong's RoMeLa group or licensing of the underlying IP represents a faster and lower-risk path than independent development.

ACADEMIC LINEAGE

The IMPASS architecture was developed at RoMeLa (UCLA) under Dr. Dennis Hong with funding from Boeing, Lockheed Martin, and the Office of Naval Research, and recognized with the 2005 ASME Freudenstein/General Motors Young Investigator Award. Two decades of published kinematic and gait analysis represents a significant head start on the locomotion engineering problem — allowing LunarMark's hardware development effort to focus on environmental qualification and science payload

integration rather than fundamental locomotion research.

Technical Differentiators

- Two-platform architecture separates broad survey (spoke wheel, fast, continuous) from precision measurement (tethered hopper, slow, deep) — maximizing data density at each scale.
- Tethered power from the crater rim eliminates the most difficult engineering challenge in PSR operations without requiring nuclear power licensing.
- The 2.5-second Earth-Moon light delay makes human-in-the-loop teleoperation feasible — unlike Mars rover operations, reducing autonomy development cost and risk.
- Ground-penetrating radar combined with neutron spectroscopy provides both structural stratigraphy and hydrogen concentration at 0-3m depth simultaneously.
- Spoke-tip sensor array on the survey platform generates continuous transect data at sub-meter resolution during transit — not point measurements at stops.
- Modular payload architecture allows sensor packages to be upgraded between missions as customer requirements evolve.

4. Business Model

Revenue Stream	Timeline	Estimated Value
NASA SBIR Phase I/II	2026-2027	\$300K-\$2M non-dilutive; funds R&D; establishes NASA relationship
CLPS Payload Contracts	2027-2029	\$5-20M per contract; NASA pays delivery cost; LunarMark retains all data rights
Data Licensing — Standard	2028+	\$500K-\$2M per dataset license; recurring for updates
Custom Survey Contracts	2029+	\$2-10M per commissioned survey; highest margin product
Data Platform Subscriptions	2029+	SaaS model; \$50K-\$500K per customer per year
Strategic Data Partnerships	2030+	Equity-for-data arrangements with mining operators; participatory upside in extraction operations

BUSINESS MODEL DESIGN

Phase 1 is deliberately asset-light. SBIR grants and CLPS contracts allow NASA to bear payload delivery cost while LunarMark operates instruments and retains all data rights. This structure — analogous to how Planet Labs used early rideshare launches — allows the company to build the most valuable asset (the dataset) before requiring significant capital.

5. Market Analysis

Market Size

- Lunar exploration technology market: \$11.4B in 2025, projected to reach \$21.4B by 2030 at 13.3% CAGR.
- Space mining market: \$1.44B in 2025, projected to reach \$4.07B by 2034 at 11.34% CAGR. Lunar mining represents 61% of that market.
- Space lander and rover market: \$2B in 2025, growing to \$5B by 2035 at 9.7% CAGR.
- Government investment: over \$31B in projected global government space exploration spending through 2034.

Target Customers

Segment	Why They Need LunarMark
NASA Artemis	South pole site characterization required before human landing; CLPS program already funds this exact category of work
ESA / JAXA / ISRO	International agencies planning south pole missions need independent data not controlled by the US government
Commercial Miners	Emerging operators need precise resource maps before committing extraction hardware capital
He-3 Operators	Interlune has a 2029 DOE delivery agreement for lunar helium-3; south pole regolith maps are operationally essential
Defense / Intelligence	Cislunar domain data is an active US national security priority; strategic data advantage has dedicated budget
Academic / Research	Surface truth data required to validate orbital and theoretical models; high volume, strong credibility-building segment

Competitive Landscape

No competitor currently offers high-resolution ground-truth PSR data as a commercial product.

- NASA VIPER: cancelled 2024 after \$450M invested. Validated the need comprehensively; left it unmet.
- Intuitive Machines IM-2 (2025): hopper attempted PSR measurements with limited data; demonstrated concept viability.
- Lunar Outpost, Astrobotic, ispace: focused on payload delivery and general mobility, not specialized PSR prospecting data products.

COMPETITIVE MOAT

First-mover data advantage compounds over time. The dataset cannot be replicated without a new surface mission. Flight heritage combined with proprietary ground-truth data — collected at sub-meter resolution by a two-platform system — creates a defensible position that grows more valuable as the lunar economy scales.

6. Milestones & Timeline

Phase	Period	Key Milestones
Phase 0: Foundation	2026	Incorporate; establish NASA and academic advisory network; initiate RoMeLa research collaboration discussion; file 2 SBIR Phase I applications; identify technical co-founder
Phase 1: R&D	2026-2027	SBIR Phase I award; prototype spoke-wheel platform in Earth analog environment; CLPS vendor relationship established; SBIR Phase II; begin flight hardware design
Phase 2: First Flight	2027-2028	First CLPS payload contract; instrument package on lunar surface; first ground-truth dataset published; Seed/Series A fundraise
Phase 3: PSR Operations	2028-2030	Rim station, spoke-wheel survey platform, and precision hopper deployed; first PSR interior measurements; full data platform launched; first commercial licensing revenue

Phase 4: Scale	2030+	Multi-site operations; strategic mining data partnerships; acquisition or IPO pathway
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7. Funding Strategy

The funding sequence is structured to minimize early equity dilution while building the technical credibility required for institutional investment:

- SBIR Phase I (\$300K, non-dilutive): target NASA SBIR topic areas covering ISRU, lunar resource prospecting, PSR instrument systems, and novel locomotion for extreme environments.
- SBIR Phase II (\$2M, non-dilutive): funds hardware prototyping and Earth-based analog testing. Spoke-wheel platform testing in Antarctic dry valleys or Atacama Desert; hopper testing in vacuum chamber facilities.
- Seed Round (\$2-4M): target Seraphim Capital, Space Capital, Bessemer, and NASA/JPL alumni angels. Funds: flight hardware build, CLPS bid preparation, key hires, RoMeLa IP licensing.
- Series A (\$15-25M): post-CLPS contract award. Funds first flight mission and full data platform development.

INVESTOR THESIS

The lunar economy is real and accelerating. The south pole is the center of gravity for all near-term activity. The data infrastructure for that economy does not yet exist. LunarMark builds it — with a two-platform robotic system that collects at both survey scale and precision scale, delivering the equivalent of the first comprehensive geological survey of a continent the world is about to move to.

8. Team Requirements

Role	Profile
CEO / Founder	Business architecture; government contracting navigation; physics or engineering background; relationship-driven approach to NASA and commercial operator go-to-market
Technical Co-Founder / CTO	Cryogenic systems, planetary robotics, or lunar instrument experience; ideally from JPL, NASA Goddard, or a CLPS

	instrument team; familiarity with spoke-wheel or passive dynamic locomotion research a strong plus
Robotics Research Partner	Collaboration with RoMeLa (UCLA) or equivalent academic group for spoke-wheel locomotion adaptation; potential for joint SBIR application
Advisor: Space Law	US Commercial Space Launch Competitiveness Act; lunar resource rights; CLPS contract structure
Advisor: NASA Procurement	Former NASA program manager or contracting officer; practical knowledge of SBIR, CLPS, and agency decision processes
Advisor: Mining / Extraction	Hard-rock or placer mining engineer; experience processing dilute ore at scale; the lunar extraction problem is more analogous to terrestrial mining than aerospace engineering

9. Risk Assessment

Risk	Likelihood	Mitigation
Starship / Artemis timeline slips	High — established pattern	Phase 0-1 SBIR revenue does not depend on lunar delivery cadence; company builds expertise regardless of schedule
Spoke-wheel platform fails dust/thermal qualification	Medium — real engineering challenge	Published IMPASS research provides strong starting point; Earth analog testing in Phase 1 de-risks before flight commitment; wheeled fallback option retained
NASA internalizes capability	Medium — current policy favors commercial	ESA, JAXA, and commercial miners are independent customers; not solely US government dependent
Lunar resource rights uncertainty	Medium — US law exists but untested	Data products are legally simpler than physical extraction; buffer period before that question is live
Better-funded competitor enters	Low near-term	Dataset irreproducible without new surface mission; two-platform architecture creates data density advantage hard to match
RoMeLa IP licensing terms	Low — academic labs generally favorable to commercial	Approach as research collaboration first; joint SBIR

	partnerships	application aligns incentives before licensing discussion
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10. Strategic Context

The China Factor

China has announced a crewed lunar south pole mission before 2030 with independent orbital infrastructure. This creates geopolitical urgency beyond pure commercial value. US and allied government customers will pay a significant premium for independent data not subject to Chinese government control. This dynamic accelerates both funding timelines and customer urgency relative to what pure commercial logic would suggest.

Analog Testing Strategy

Prior to lunar deployment, both platforms must be validated in Earth environments approximating south pole conditions. The Atacama Desert, Antarctic dry valleys, and volcanic lava tubes in Iceland and Hawaii are established NASA and ESA lunar analogs. The spoke-wheel platform is particularly well suited to analog testing — its Earth-gravity performance can be characterized first, with gait parameters then adjusted for lunar gravity through simulation and incremental testing. Analog campaigns generate publishable data, build research community relationships, and create compelling investor and customer materials.

Helium-3 Market Emergence

In May 2025, Seattle-based startup Interlune announced a Department of Energy agreement to deliver helium-3 from the lunar surface to Earth by 2029 for quantum computing and medical imaging. He-3 is present in lunar regolith across the south pole from billions of years of solar wind bombardment. LunarMark's prospecting data — particularly the continuous transect surveys enabled by the spoke-wheel platform — would be directly valuable to He-3 extraction planning across wide areas.

International Customer Base

NASA is the natural first customer but the addressable market extends significantly beyond US procurement. ESA, JAXA, and ISRO all have active south pole programs and budgets for independent data. International customers provide meaningful political risk diversification and should be developed in parallel with domestic customer relationships from the earliest stage.

11. Go-To-Market Sequence

Phase 1: Establish Domain Credibility (2026)

- Register with NASA CLPS vendor program and attend CLPS industry days.
- Contact RoMeLa / Dr. Dennis Hong regarding research collaboration on lunar adaptation of IMPASS locomotion architecture.
- Apply to NASA SBIR solicitation; relevant topic areas include ISRU, lunar resource prospecting, PSR instrumentation, and novel surface mobility for extreme environments.
- Engage Colorado School of Mines Space Resources Program — leading academic center for space resource utilization research.
- Attend Space Resources Roundtable, Lunar Surface Innovation Consortium (LSIC) workshops, AIAA SPACE Forum.
- Publish technical concept note on LunarMark two-platform data collection architecture for circulation within the lunar science and operations community.

Phase 2: First Revenue (2026-2027)

- SBIR Phase I award converts to non-dilutive R&D funding and validates NASA relationship.
- Joint SBIR application with RoMeLa academic partner accelerates locomotion development and adds institutional credibility.
- First paying engagement: pre-mission site characterization analysis commissioned by a CLPS lander operator or commercial mining company.

Phase 3: Flight Heritage (2027-2029)

- First CLPS payload contract; instrument package manifested on a commercial lander mission to the south pole.
- First ground-truth dataset published — establishes LunarMark as the source of record for south pole surface data.
- Seed round closed on the basis of demonstrated flight heritage and active customer contracts.

Summary

LunarMark addresses the single most important data gap in the emerging lunar economy: high-resolution, ground-truth characterization of the resources that make sustained human presence at the lunar south pole viable. The two-platform robotic architecture — an IMPASS-derived

active spoke wheel platform for rapid broad-area survey combined with a precision tethered hopper for PSR interior access — delivers data density and coverage that no single-platform design can match.

The business is differentiated by technical specificity, protected by the irreproducibility of first-surface data, and structured to generate revenue at each stage before requiring the next round of capital. The locomotion architecture draws on two decades of published academic research, reducing development risk while providing a clear path to a research collaboration that strengthens both the technical and institutional credibility of the program.

INVESTMENT THESIS

Every entity planning to operate at the lunar south pole needs the data LunarMark produces. The dataset cannot be bought, only built. The company that builds it first — with the most comprehensive collection architecture — owns the infrastructure layer of the lunar economy for the next decade.